

Day 7 — Limits of the Method

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: recognise when traditional long multiplication is faster than the sūtra method.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthaji”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Pose the question.** “*When IS the $\times 11$ (or near-base) method genuinely faster than traditional long multiplication? When is it the SAME or SLOWER?*”
2. **Categorise problem types.**
 - 2-digit $\times 11$ — much faster.
 - 3-digit $\times 11$ — somewhat faster.
 - 2-digit $\times 12$ –19 — comparable.
 - 2-digit $\times 23$ (far from base) — long multiplication is faster.
 - Long multi-digit products — long multiplication or scientific notation is best.
3. **Honest framing.** Tirthaji’s sūtras are not a universal speed-up. They are a *toolbox of mental shortcuts for specific patterns*. The skill is recognising when to use which tool.
4. **Reference.** Modern mathematicians (S.G. Dani, 1993; Kim Plofker, *Mathematics in India*, 2009) have written critically about the “Vedic” claim and pedagogically about the methods. Read Dani’s *Frontline* article excerpt aloud (one paragraph).

5. **Reflection.** Each student writes 3 sentences: which method-types are useful, what limits are, and where standard arithmetic remains best.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Limits of the Method.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).